Problem statements

Introduction

The problem statements given in this section are to be taken up by groups of 5 students. You, as a group, need to submit a report and make a presentation towards the end of the semester on these problems. **10 minutes** will be allotted for each presentation, followed by a discussion for **5 minutes**. The project report carries **10 marks** towards your grade. The presentation and the subsequent discussion carry **10 marks** towards your final grade.

Your project report and presentation must include the following:

- Contribution of each group member
- Details of the reaction mechanism used.
- Screenshots of CHEMKIN panels specific to your problem.
- Results

Group formation

Below is the complete allocation table for all 16 groups. Students randomly assigned are indicated with an asterisk (*).

Group	Member 1	Member 2	Member 3	Member 4	Member 5
Group 1	23MF10002 (Anirudh Agrawal)	23ME31027 (Saptak Hazra)	23ME31048 (Kunal Mandal)	23ME31044 * (Sohan Datta)	22ME6001 * (Ankit Singh)
Group 2	23ME10058 (Piyush Ranjan)	23ME10032 (Harshit Meena)	23ME31014 * (Krish Kumar)	21ME10018 * (Ayush Kumar Barda)	22ME36018 * (Himanshu Sahu)
Group 3	22ME36008 (Shivam Kumar Singh)	23ME31031 (Shubham Kumar)	23ME33009 (Siddharth Jain)	23ME31011 * (Kappala Abhishek Varam)	—
Group 4	23MF10007 (Aryan Malhotra)	23MF10026 (Naavya Modi)	23MF10041 (Tanisha Jaiswal)	23ME10078 (Sharvari Rajesh Karhe)	23ME31047 * (Sailor Dhruv Jayeshbhai)
Group 5	23ME10109 (Samarth Jauhal)	23ME31051 (Vanshika Agarwal)	23ME10115 (Gaurav Singh)	25ME72P03 (Ashutosh Kumar)	23MF10013 (Harsh Vibhor Sharma)
Group 6	23ME36001 (Aditya Prakash Gupta)	23ME36002 (Ashish Meena)	23ME36005 (Soham Mukherjee)	23MF10027 (Naveen Saini)	23ME31032 * (Swayam Kiran Chavan)
Group 7	23ME10120 (Venugopal Venkat)	23ME33010 (Suvam Debnath)	23ME33013 (Sidhart Sharma)	23ME33014 (Soumadip Majumder)	23IM10014 (Jash Nikhil Mehta)
Group 8	23ME10104 (Sritoma Nandan)	23ME10035 (Isha Singhal)	23ME31036 (Ishan Rai)	23ME10047 (Mantra Malpani)	23ME31012 (Killi Sruthi)

Group formation

Below is the complete allocation table for all 16 groups. Students randomly assigned are indicated with an asterisk (*).

Group	Member 1	Member 2	Member 3	Member 4	Member 5
Group 9	23ME10048 (Mauryavardhan Singh)	23ME10049 (Mayank Nagar)	23ME10043 (Krityapriya Bhaumik)	23ME31037 (Hardik Mahawar)	23ME10011 (Apurba Kumar Show)
Group 10	23ME10089 (Sudeep Dileep Kumar Prajapati)	23ME10121 (Kreesh Vijay Modi)	23ME10096 (Vedansh Patel)	23ME10107 (Prem Agarwal)	23ME10106 (Sahil Kumar Gupta)
Group 11	23ME10060 (Pranjali Khatua)	23ME10066 (Raksha Gupta)	23ME10038 (Kanchan Potdar)	23ME10053 (Nagineni Lakshmipravallika)	23ME10074 (Saloni)
Group 12	22MF3IM32 (Subhajit Mukhopadhyay)	22MF10002 (Adwait Uday Khochare)	22MF3IM12 (Mudavath Simhachal)	23ME31022 * (Rudranarayan Pradhan)	25ME10022 * (Diya Arjun Jadhav)
Group 13	21ME31026 (Karan Nag)	23ME10124 * (Pitambar Mukherjee)	23ME10029 * (Gokul R)	23MF10004 * (Anubhab Sharma)	20ME36008 * (Palla Sri Harshavardhan)
Group 14	23ME10009 (Anmol Dureja)	23ME10113 (Videh Jha)	23MF10014 (Hitarth Pritesh Vora)	23MF10035 (Sanghmitra Rai)	23ME10081 (Siddhant Yadav)
Group 15	26ME62R04 (Areet Biswas)	26ME62R11 (Veerabhadra Sri Manisankar Kalepu)	26ME62R19 (Roopchand Chowdary Sunkara)	23ME10042 * (Krishnav Anand Agarwal)	25ME62D01 * (Umesh Kala)
Group 16	25AG92R06 (Jaydev Mahto)	23ME10097 (Vishal Kumar)	23ME31024 (Salodkar Atharv)	23ME10101 (Ayush Singh)	—

Problem allocation

Group 1 will work on Problem 1, Group 2 will work on Problem 2, and so on.

Submission Deadline

The project reports must be uploaded to the following link by 10th November, 2026.

Link: https://iitkgpacin-my.sharepoint.com/:f:/g/personal/pabitrabadhuk_mech_iitkgp_ac_in/IgAuoET4n7k-T4X8gq3jx0bqAW7FbAQVbcEzfa4XxBCUeA4?e=hGtPZ3

Group Project Presentation Schedule:

Groups 1 to 8: 3rd November, 2026. Venue: In the classroom during class hours (4 pm to 6 pm).
Groups 9 to 16: 10th November, 2026. Venue: In the classroom during class hours (4 pm to 6 pm).

You may reach out to the TAs for help with Ansys CHEMKIN installation and other aspects.

Groups 1 to 8: Naveen Kumar (naveenkr00077925@kgpian.iitkgp.ac.in)
Groups 9 to 16: Rohit Roy (ROYROHIT.24@KGPIAN.IITKGP.AC.IN)

Problem 1

Use the sample file titled **equilibrium__gas.ckprj** available in the **samples2025R2** folder as the starting point of this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project. Calculate the adiabatic flame temperature and the equilibrium mole fractions of CO, CO_2, and H_2O for the following conditions.

(a) Fuel: iso-octane, oxidizer: air, const. volume, $T_0 = 800$ K, $P_0 = 10$ atm, $\phi = 0.5$ to 2.0.
(b) Fuel: gasoline [blend of iso-octane (86%, by moles) and n-heptane (14%, by moles)], oxidizer: air, const. volume, $T_0 = 800$ K, $P_0 = 10$ atm, $\phi = 0.5$ to 2.0.
(c) Fuel: blend of gasoline (80% by volume) and ethanol (20% by volume), oxidizer: air, const. volume, $T_0 = 800$ K, $P_0 = 10$ atm, $\phi = 0.5$ to 2.0.
(d) Fuel: iso-octane, oxidizer: air, const. volume, $T_0 = 800$ K, $P_0 = 1$ atm, $\phi = 0.5$ to 2.0.
(e) Fuel: iso-octane, oxidizer: air, const. volume, $T_0 = 300$ K, $P_0 = 10$ atm, $\phi = 0.5$ to 2.0.

Clearly bring out the influence of fuel composition, T_0, P_0, and equivalence ratio on the adiabatic flame temperature and equilibrium mole fractions of CO, CO_2, and H_2O. **Explain your observation from first principles.**

Problem 2

Use the sample file titled **closed_homogeneous__ignition_delay.ckprj** available in the **samples2025R2** folder as the starting point of this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project. Calculate the **auto-ignition delay** of the following mixtures.

(a) Fuel: iso-octane, oxidizer: air, const. volume, volume = 50 cm^3, T_0 = 300 K, P_0 = 1 atm, ϕ = 1.
(b) Fuel: iso-octane, oxidizer: air, const. volume, volume = 10 cm^3, T_0 = 1500 K, P_0 = 1 atm, ϕ = 1.
(c) Fuel: iso-octane, oxidizer: air, const. volume, volume = 50 cm^3, T_0 = 1500 K, P_0 = 10 atm, ϕ = 1.
(d) Fuel: iso-octane, oxidizer: air, const. volume, volume = 50 cm^3, T_0 = 1500 K, P_0 = 10 atm, ϕ = 0.9.
(e) Fuel: n-heptane, oxidizer: air, const. volume, volume = 50 cm^3, T_0 = 1500 K, P_0 = 10 atm, ϕ = 1.
(f) Fuel: iso-octane, oxidizer: air + 10% CO_2 (by volume), const. volume, volume = 50 cm^3, T_0 = 1000 K, P_0 = 10 atm, ϕ = 1.

Clearly bring out the influence of fuel, oxidizer, reactor volume, T_0, P_0, and equivalence ratio on the ignition delay. **Explain your observation from first principles.**

Problem 3

Use the sample file titled **closed_homogeneous__ignition_delay.ckprj** available in the **samples2025R2** folder as the starting point of this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project. Calculate the **auto-ignition delay** of the following mixtures.

(a) Fuel: n-heptane, oxidizer: air, const. pressure, $T_0 = 300$ K, $P_0 = 1$ atm, $\phi = 1$.
(b) Fuel: n-heptane, oxidizer: air, const. pressure, $T_0 = 1500$ K, $P_0 = 1$ atm, $\phi = 1$.
(c) Fuel: n-heptane, oxidizer: air, const. pressure, $T_0 = 1500$ K, $P_0 = 10$ atm, $\phi = 1$.
(d) Fuel: n-heptane, oxidizer: air, const. pressure, $T_0 = 1500$ K, $P_0 = 10$ atm, $\phi = 0.9$.
(e) Fuel: iso-octane, oxidizer: air, const. pressure, $T_0 = 1500$ K, $P_0 = 10$ atm, $\phi = 1$.
(f) Fuel: n-heptane, oxidizer: air + 10% CO_2 (by volume), const. pressure, $T_0 = 1000$ K, $P_0 = 10$ atm, $\phi = 1$.

Clearly bring out the influence of fuel, oxidizer, reactor volume, T_0, P_0, and equivalence ratio on the ignition delay. **Explain your observation from first principles.**

Problem 4

Use the sample file titled **closed_homogeneous__ignition_delay.ckprj** available in the **samples2025R2** folder as the starting point of this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project. Calculate the **auto-ignition delay** of the following mixtures.

(a) Fuel: n-heptane, oxidizer: air, const. pressure, T_0 = 1000 K – 2000 K, P_0 = 10 atm, ϕ = 1.
(b) Fuel: iso-octane, oxidizer: air, const. pressure, T_0 = 1000 K – 2000 K, P_0 = 10 atm, ϕ = 1.
(c) Fuel: gasoline [blend of iso-octane (86%, by moles) and n-heptane (14%, by moles)], oxidizer: air, const. pressure, T_0 = 1000 K – 2000 K, P_0 = 10 atm, ϕ = 1.
(d) Fuel: diesel [blend of $C_{16}H_{34}$ (90%, by moles) and C_7H_8 (10%, by moles)], oxidizer: air, const. pressure, T_0 = 1000 K – 2000 K, P_0 = 10 atm, ϕ = 1.
(e) Fuel: blend of gasoline (80% by volume) and ethanol (20% by volume), oxidizer: air, const. pressure, T_0 = 1000 K – 2000 K, P_0 = 10 atm, ϕ = 1.

Clearly bring out the influence of fuel and T_0 on the ignition delay. **Explain your observation from first principles.**

Problem 5

Use the sample file titled **ic_engine__nbutane.ckprj** available in the **samples2025R2** folder as the starting point of this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project.

Use the following Reactant Mixture composition.
Fuel: Iso-octane (ic8h18)
Oxidizer: Air (21% o2, 79% n2 by moles)

Start with the default values in the sample. Study the influence of engine speed (vary from 500 to 3000 RPM), engine compression ratio (vary from 6 to 20), and equivalence ratio (vary from 0.7 to 1.3) on the torque, power, and specific fuel consumption. Obtain the **global optimal values** of these input parameters over the entire parameter space that (a) maximize the torque (IVC-EVO), (b) maximize the power (IVC-EVO), and (c) minimize the specific fuel consumption (IVC-EVO).
Explain your observation from first principles.
Compare the cycle efficiency of the Otto cycle for the same compression ratio. Explain the differences.
Explain the isolated influence of each input parameter (with all other inputs fixed).

Problem 6

Use the sample file titled **ic_engine__nbutane.ckprj** available in the **samples2025R2** folder as the starting point of this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project.

Use the following Reactant Mixture composition.
Fuel: n-heptane (nc7h16)
Oxidizer: Air (21% O2, 79% N2 by moles)

Start with the default values in the sample. Study the influence of engine speed (vary from 500 to 3000 RPM), engine compression ratio (vary from 6 to 20), and equivalence ratio (vary from 0.7 to 1.3) on the torque, power, and specific fuel consumption. Obtain the **global optimal values** of these input parameters over the entire parameter space that (a) maximize the torque (IVC-EVO), (b) maximize the power (IVC-EVO), and (c) minimize the specific fuel consumption (IVC-EVO).
Explain your observation from first principles.
Compare the cycle efficiency of the Otto cycle for the same compression ratio. Explain the differences.
Explain the isolated influence of each input parameter (with all other inputs fixed).

Problem 7

Use the sample file titled **ic_engine__nbutane.ckprj** available in the **samples2025R2** folder as the starting point of this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project.

Use the following Reactant Mixture composition.
Fuel: Ethanol (c2h5oh)
Oxidizer: Air (21% o2, 79% n2 by moles)

Start with the default values in the sample. Study the influence of engine speed (vary from 500 to 3000 RPM), engine compression ratio (vary from 6 to 20), and equivalence ratio (vary from 0.7 to 1.3) on the torque, power, and specific fuel consumption. Obtain the **global optimal values** of these input parameters over the entire parameter space that (a) maximize the torque (IVC-EVO), (b) maximize the power (IVC-EVO), and (c) minimize the specific fuel consumption (IVC-EVO).
Explain your observation from first principles.
Compare the cycle efficiency of the Otto cycle for the same compression ratio. Explain the differences.
Explain the isolated influence of each input parameter (with all other inputs fixed).

Problem 8

Use the sample file titled **ic_engine__nbutane.ckprj** available in the **samples2025R2** folder as the starting point of this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project.

Use the following Reactant Mixture composition.
Fuel: Methanol (ch3oh)
Oxidizer: Air (21% o2, 79% n2 by moles)

Start with the default values in the sample. Study the influence of engine speed (vary from 500 to 3000 RPM), engine compression ratio (vary from 6 to 20), and equivalence ratio (vary from 0.7 to 1.3) on the torque, power, and specific fuel consumption. Obtain the **global optimal values** of these input parameters over the entire parameter space that (a) maximize the torque (IVC-EVO), (b) maximize the power (IVC-EVO), and (c) minimize the specific fuel consumption (IVC-EVO).
Explain your observation from first principles.
Compare the cycle efficiency of the Otto cycle for the same compression ratio. Explain the differences.
Explain the isolated influence of each input parameter (with all other inputs fixed).

Problem 9

Use the sample file titled **ic_engine__SI_engine.ckprj** available in the **samples2025R2** folder as the starting point for this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project.

Use the following Reactant Mixture composition.
Fuel: Iso-octane (ic8h18)
Oxidizer: Air (21% O2, 79% N2 by moles)

Start with the default values in the sample. Study the influence of engine speed (vary from 500 to 3000 RPM), start of combustion crank angle (-15° to -5°), and burn duration (10° to 50°) on the torque, power, and thermal efficiency. Obtain the **global optimal values** of these input parameters over the entire parameter space that (a) maximize the torque (IVC-EVO), (b) maximize the power (IVC-EVO), and (c) minimize the specific fuel consumption (IVC-EVO).

Explain your observation from first principles.

Compare the cycle efficiency of the Otto cycle for the same compression ratio. Explain the differences.

Explain the isolated influence of each input parameter (with all other inputs fixed).

Problem 10

Use the sample file titled **ic_engine__SI_engine.ckprj** available in the **samples2025R2** folder as the starting point for this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project.

Use the following Reactant Mixture composition.
Fuel: n-heptane (nc7h16)
Oxidizer: Air (21% o2, 79% n2 by moles)

Start with the default values in the sample. Study the influence of engine speed (vary from 500 to 3000 RPM), start of combustion crank angle (-15° to -5°), and burn duration (10° to 50°) on the torque, power, and thermal efficiency. Obtain the **global optimal values** of these input parameters over the entire parameter space that (a) maximize the torque (IVC-EVO), (b) maximize the power (IVC-EVO), and (c) minimize the specific fuel consumption (IVC-EVO).

Explain your observation from first principles.

Compare the cycle efficiency of the Otto cycle for the same compression ratio. Explain the differences.

Explain the isolated influence of each input parameter (with all other inputs fixed).

Problem 11

Use the sample file titled **ic_engine__SI_engine.ckprj** available in the **samples2025R2** folder as the starting point for this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project.

Use the following Reactant Mixture composition.
Fuel: Ethanol (c2h5oh)
Oxidizer: Air (21% o2, 79% n2 by moles)

Start with the default values in the sample. Study the influence of engine speed (vary from 500 to 3000 RPM), start of combustion crank angle (-15° to -5°), and burn duration (10° to 50°) on the torque, power, and thermal efficiency. Obtain the **global optimal values** of these input parameters over the entire parameter space that (a) maximize the torque (IVC-EVO), (b) maximize the power (IVC-EVO), and (c) minimize the specific fuel consumption (IVC-EVO).

Explain your observation from first principles.

Compare the cycle efficiency of the Otto cycle for the same compression ratio. Explain the differences.

Explain the isolated influence of each input parameter (with all other inputs fixed).

Problem 12

Use the sample file titled **ic_engine__SI_engine.ckprj** available in the **samples2025R2** folder as the starting point for this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project.

Use the following Reactant Mixture composition.
Fuel: Methanol (ch3oh)
Oxidizer: Air (21% o2, 79% n2 by moles)

Start with the default values in the sample. Study the influence of engine speed (vary from 500 to 3000 RPM), start of combustion crank angle (-15° to -5°), and burn duration (10° to 50°) on the torque, power, and thermal efficiency. Obtain the **global optimal values** of these input parameters over the entire parameter space that (a) maximize the torque (IVC-EVO), (b) maximize the power (IVC-EVO), and (c) minimize the specific fuel consumption (IVC-EVO).
Explain your observation from first principles.
Compare the cycle efficiency of the Otto cycle for the same compression ratio. Explain the differences.
Explain the isolated influence of each input parameter (with all other inputs fixed).

Problem 13

Use the sample file titled **ic_engine__SI_engine.ckprj** available in the **samples2025R2** folder as the starting point for this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project.

Use the following Reactant Mixture composition.
Fuel: Iso-octane (ic8h18)
Oxidizer: Air (21% O2, 79% N2 by moles)

Start with the default values in the sample. Study the influence of the Wiebe function parameters (n and b) on the torque, power, and thermal efficiency (all IVC-EVO). Report the optimal Wiebe function parameters (n and b) for various start of combustion crank angle (-15° to -5°), and burn duration (10° to 50°).

Explain your observation from first principles.

Compare the cycle efficiency of the Otto cycle for the same compression ratio. Explain the differences.

Explain the isolated influence of each input parameter (with all other inputs fixed).

Problem 14

Use the sample file titled **ic_engine__SI_engine.ckprj** available in the **samples2025R2** folder as the starting point for this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project.

Use the following Reactant Mixture composition.
Fuel: n-heptane (nc7h16)
Oxidizer: Air (21% o2, 79% n2 by moles)

Start with the default values in the sample. Study the influence of the Wiebe function parameters (n and b) on the torque, power, and thermal efficiency (all IVC-EVO). Report the optimal Wiebe function parameters (n and b) for various start of combustion crank angle (-15° to -5°), and burn duration (10° to 50°).

Explain your observation from first principles.

Compare the cycle efficiency of the Otto cycle for the same compression ratio. Explain the differences.

Explain the isolated influence of each input parameter (with all other inputs fixed).

Problem 15

Use the sample file titled **ic_engine__SI_engine.ckprj** available in the **samples2025R2** folder as the starting point for this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project.

Use the following Reactant Mixture composition.
Fuel: Ethanol (c2h5oh)
Oxidizer: Air (21% o2, 79% n2 by moles)

Start with the default values in the sample. Study the influence of the Wiebe function parameters (n and b) on the torque, power, and thermal efficiency (all IVC-EVO). Report the optimal Wiebe function parameters (n and b) for various start of combustion crank angle (-15° to -5°), and burn duration (10° to 50°).

Explain your observation from first principles.

Compare the cycle efficiency of the Otto cycle for the same compression ratio. Explain the differences.

Explain the isolated influence of each input parameter (with all other inputs fixed).

Problem 16

Use the sample file titled **ic_engine__SI_engine.ckprj** available in the **samples2025R2** folder as the starting point for this problem. Feed the mechanism, thermodynamics, and transport files from the Isooctane_nHeptane_Ethanol_Methanol folder into this project.

Use the following Reactant Mixture composition.
Fuel: Methanol (ch3oh)
Oxidizer: Air (21% o2, 79% n2 by moles)

Start with the default values in the sample. Study the influence of the Wiebe function parameters (n and b) on the torque, power, and thermal efficiency (all IVC-EVO). Report the optimal Wiebe function parameters (n and b) for various start of combustion crank angle (-15° to -5°), and burn duration (10° to 50°).

Explain your observation from first principles.

Compare the cycle efficiency of the Otto cycle for the same compression ratio. Explain the differences.

Explain the isolated influence of each input parameter (with all other inputs fixed).